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APTITUDE QUESTION BANK

Quantitative Aptitude

Placement preparation for IT company hiring tests

Where the Syllabus Ends and Reality Begins

150	Q1 – Q50	Q51 – Q100	Q101 – Q150
Multiple-choice questions	Beginner	Intermediate	Advanced

About this section

Quantitative Aptitude checks how quickly and accurately you work with numbers. Hiring tests at companies such as TCS, Infosys, Wipro, Accenture and Cognizant regularly include percentages, profit and loss, ratios, averages, time and work, time-speed-distance, interest, number systems, mixtures, permutations and probability. Learn the basic formula for each topic, then practise shortcuts so that each question takes about a minute.

How to use this question bank

- Part A has all 150 questions, grouped by difficulty. Attempt them on your own first, with a timer if you can (about 1 minute per question).
- Part B is the answer key. Check your answers only after finishing a full level.
- Part C explains how to solve every question in simple steps. Read the solution even for the ones you got right — it often shows a faster method.
- Mark the questions you got wrong and attempt them again after two or three days.

PART A

Questions

BEGINNER Questions 1 – 50

Q1. What is 25% of 480?

A) 125
B) 100
C) 140
D) 120

Q2. A shirt costs ₹800 and is sold for ₹920. What is the profit percentage?

A) 18%
B) 15%
C) 20%
D) 12%

Q3. What is the average of 12, 18, 24, 30 and 36?

A) 24
B) 22
C) 30
D) 26

Q4. A car travels 180 km in 3 hours. What is its speed?

A) 45 km/h
B) 54 km/h
C) 60 km/h
D) 50 km/h

Q5. What is the simple interest on ₹5,000 at 8% per year for 3 years?

A) ₹1,200
B) ₹1,500
C) ₹1,000
D) ₹1,800

Q6. Divide ₹720 between A and B in the ratio 5 : 3. What is A's share?

A) ₹400
B) ₹450
C) ₹480
D) ₹270

Q7. What is the LCM of 12 and 18?

A) 54
B) 72
C) 6
D) 36

Q8. What is the HCF of 48 and 72?

A) 36
B) 12
C) 24
D) 18

Q9. If 8 pens cost ₹96, how much do 15 pens cost?

A) ₹180
B) ₹170
C) ₹192
D) ₹160

Q10. A number increased by 20% becomes 360. What is the number?

A) 320
B) 340
C) 300
D) 288

Q11. What is the value of $15^2 - 10^2$?

A) 100
B) 125
C) 25
D) 50

Q12. A can finish a job in 10 days. What part of the job does A finish in 1 day?

A) 10
B) 1/5
C) 1/20
D) 1/10

Q13. The price of an item drops from ₹250 to ₹200. What is the percentage decrease?

A) 15%
B) 50%
C) 25%
D) 20%

Q14. What is 3/5 of 250?

A) 150
B) 100
C) 125
D) 175

Q15. Find the next number: 2, 6, 12, 20, 30, ?

A) 36
B) 42
C) 44
D) 40

Q16. The sum of two numbers is 50 and their difference is 10. What is the larger number?

- **A)** 20
- **B)** 25
- **C)** 30
- **D)** 35

Q17. How many minutes are there in 3.5 hours?

- **A)** 200
- **B)** 180
- **C)** 210
- **D)** 350

Q18. A train 120 m long passes a pole in 6 seconds. What is its speed in m/s?

- **A)** 20 m/s
- **B)** 18 m/s
- **C)** 24 m/s
- **D)** 12 m/s

Q19. Convert 72 km/h into m/s.

- **A)** 72 m/s
- **B)** 18 m/s
- **C)** 25 m/s
- **D)** 20 m/s

Q20. What is 0.25 written as a fraction in lowest terms?

- **A)** 2/5
- **B)** 1/4
- **C)** 1/25
- **D)** 1/2

Q21. A shopkeeper buys a toy for ₹400 and sells it at a loss of 10%. What is the selling price?

- **A)** ₹360
- **B)** ₹440
- **C)** ₹340
- **D)** ₹380

Q22. The ratio of boys to girls in a class is 3 : 2. If there are 18 boys, how many girls are there?

- **A)** 15
- **B)** 27
- **C)** 12
- **D)** 10

Q23. What is the square root of 1,296?

- **A)** 38
- **B)** 36
- **C)** 46
- **D)** 34

Q24. Which of these numbers is a prime number?

A) 51
B) 39
C) 57
D) 29

Q25. A person walks at 5 km/h. How far does he walk in 2 hours 30 minutes?

A) 15 km
B) 11 km
C) 10 km
D) 12.5 km

Q26. The average age of 4 friends is 20 years. What is the total of their ages?

A) 84 years
B) 60 years
C) 80 years
D) 24 years

Q27. What is 40% of 60% of 500?

A) 100
B) 120
C) 200
D) 300

Q28. If x + 7 = 19, what is the value of 2x?

A) 24
B) 26
C) 38
D) 12

Q29. How many numbers between 1 and 50 (inclusive) are divisible by 7?

A) 7
B) 8
C) 5
D) 6

Q30. What is the perimeter of a rectangle with length 12 cm and breadth 8 cm?

A) 20 cm
B) 48 cm
C) 96 cm
D) 40 cm

Q31. What is the area of a square whose side is 9 cm?

A) 18 sq cm
B) 81 sq cm
C) 72 sq cm
D) 36 sq cm

Q32. A sum of money doubles in 10 years at simple interest. What is the rate of interest?

A) 5%
B) 20%
C) 10%
D) 12%

Q33. What is the value of (64)^(1/2) + (27)^(1/3)?

A) 10
B) 12
C) 17
D) 11

Q34. If 5 workers build a wall in 12 days, how many days will 10 workers take (same speed)?

A) 24
B) 6
C) 10
D) 8

Q35. What is 1/2 + 1/3 + 1/6?

A) 1
B) 2/3
C) 1/2
D) 5/6

Q36. A bag has 3 red and 5 blue balls. One ball is picked at random. What is the probability that it is red?

A) 5/8
B) 3/5
C) 3/8
D) 1/3

Q37. Which is the smallest: 0.5, 0.45, 0.405, 0.54?

A) 0.405
B) 0.5
C) 0.45
D) 0.54

Q38. What is the cost price if an article sold for ₹660 gives a profit of 10%?

A) ₹594
B) ₹650
C) ₹600
D) ₹726

Q39. A tank fills in 6 hours with one pipe. How much of the tank fills in 2 hours?

A) 2/3
B) 1/3
C) 1/6
D) 1/2

Q40. What is 7 × 8 + 12 ÷ 4 − 5?

A) 15
B) 59
C) 48
D) 54

Q41. A student scored 45 out of 60. What is the percentage?

A) 70%
B) 65%
C) 80%
D) 75%

Q42. The sum of three consecutive numbers is 72. What is the largest of them?

A) 26
B) 25
C) 23
D) 24

Q43. How many seconds are there in one day?

A) 864,000
B) 8,640
C) 86,400
D) 3,600

Q44. What is the remainder when 100 is divided by 7?

A) 2
B) 4
C) 1
D) 3

Q45. A man's monthly salary is ₹30,000. He saves 15% of it. How much does he save?

A) ₹4,500
B) ₹5,500
C) ₹4,000
D) ₹3,000

Q46. If the cost of 1 dozen bananas is ₹48, what is the cost of 1 banana?

A) ₹5
B) ₹6
C) ₹4
D) ₹3

Q47. What is the next number: 3, 9, 27, 81, ?

A) 324
B) 243
C) 162
D) 189

Q48. The angles of a triangle are in the ratio 1 : 2 : 3. What is the largest angle?

- **A)** 120°
- **B)** 100°
- **C)** 60°
- **D)** 90°

Q49. Express 3/8 as a percentage.

- **A)** 30%
- **B)** 38%
- **C)** 37.5%
- **D)** 35%

Q50. A bus leaves at 9:45 AM and reaches at 1:15 PM. How long is the journey?

- **A)** 3 hours 30 minutes
- **B)** 4 hours 30 minutes
- **C)** 3 hours 15 minutes
- **D)** 3 hours 45 minutes

INTERMEDIATE Questions 51 – 100

Q51. A can do a work in 12 days and B can do it in 18 days. In how many days can they finish it together?

A) 15 days
B) 7.2 days
C) 6 days
D) 8 days

Q52. A train 200 m long crosses a 300 m long platform in 25 seconds. What is the speed of the train in km/h?

A) 80 km/h
B) 60 km/h
C) 54 km/h
D) 72 km/h

Q53. Find the compound interest on ₹10,000 at 10% per year for 2 years (compounded yearly).

A) ₹2,100
B) ₹1,100
C) ₹2,000
D) ₹2,200

Q54. A shopkeeper marks an item 40% above cost and gives a 20% discount. What is his profit percentage?

A) 10%
B) 12%
C) 20%
D) 16%

Q55. The average of 5 numbers is 40. If one number is removed, the average becomes 38. What is the removed number?

A) 50
B) 42
C) 48
D) 40

Q56. A boat goes 24 km downstream in 2 hours and returns upstream in 4 hours. What is the speed of the stream?

A) 6 km/h
B) 9 km/h
C) 2 km/h
D) 3 km/h

Q57. The ratio of A's to B's salary is 4 : 5. If A's salary increases by 25% and B's by 20%, what is the new ratio?

A) 5 : 6
B) 1 : 1
C) 4 : 5
D) 6 : 5

Q58. Two pipes fill a tank in 20 and 30 minutes. A third pipe empties it in 15 minutes. If all three are opened together, how long will it take to fill the tank?

A) 30 minutes
B) Never fills
C) 60 minutes
D) 45 minutes

Q59. How many ways can the letters of the word 'APPLE' be arranged?

A) 120
B) 24
C) 30
D) 60

Q60. A sum becomes ₹6,600 in 3 years and ₹7,200 in 5 years at simple interest. What is the principal?

A) ₹6,000
B) ₹5,700
C) ₹5,400
D) ₹5,000

Q61. Two dice are thrown. What is the probability that the sum is 8?

A) 1/6
B) 5/36
C) 1/9
D) 7/36

Q62. In what ratio must rice at ₹40/kg be mixed with rice at ₹55/kg to get a mixture worth ₹45/kg?

A) 1 : 2
B) 2 : 3
C) 3 : 2
D) 2 : 1

Q63. A person covers half a journey at 40 km/h and the other half at 60 km/h. What is his average speed?

A) 48 km/h
B) 50 km/h
C) 52 km/h
D) 45 km/h

Q64. The population of a town is 50,000. It grows by 10% every year. What will it be after 2 years?

A) 55,000
B) 60,000
C) 60,500
D) 61,000

Q65. What is the unit digit of 7^45?

A) 1
B) 7
C) 9
D) 3

Q66. The price of sugar increases by 25%. By how much percent must a family reduce consumption so that spending stays the same?

A) 25%
B) 15%
C) 30%
D) 20%

Q67. A and B start a business with ₹30,000 and ₹45,000. After a year the profit is ₹15,000. What is B's share?

A) ₹6,000
B) ₹10,000
C) ₹9,000
D) ₹7,500

Q68. A clock shows 3:40. What is the angle between the hour and minute hands?

A) 130°
B) 150°
C) 120°
D) 140°

Q69. The difference between CI and SI on a sum for 2 years at 5% per year is ₹25. What is the sum?

A) ₹12,500
B) ₹10,000
C) ₹8,000
D) ₹5,000

Q70. A man sells two watches for ₹990 each. On one he gains 10% and on the other he loses 10%. What is his overall result?

A) 1% gain
B) 2% loss
C) 1% loss
D) No profit no loss

Q71. How many 3-digit numbers can be formed using the digits 1, 2, 3, 4, 5 without repetition?

A) 20
B) 120
C) 125
D) 60

Q72. A train running at 54 km/h crosses a man walking at 6 km/h in the same direction in 18 seconds. What is the length of the train?

A) 240 m
B) 270 m
C) 180 m
D) 300 m

Q73. The ages of A and B are in the ratio 3 : 5. After 6 years the ratio will be 2 : 3. What is A's present age?

A) 12 years
B) 15 years
C) 18 years
D) 24 years

Q74. If 15 men can do a job in 20 days working 8 hours a day, how many days will 20 men take working 6 hours a day?

A) 20 days
B) 15 days
C) 18 days
D) 25 days

Q75. What is the least number which, when divided by 6, 8 and 12, leaves a remainder 5 in each case?

A) 24
B) 29
C) 53
D) 35

Q76. A sells an item to B at 20% profit, and B sells it to C at 25% profit. If C pays ₹600, what did A pay?

A) ₹360
B) ₹450
C) ₹480
D) ₹400

Q77. Out of 100 students, 60 like tea, 50 like coffee and 20 like both. How many like neither?

A) 30
B) 20
C) 0
D) 10

Q78. A number when divided by 5 leaves remainder 3. What is the remainder when the square of the number is divided by 5?

A) 9
B) 3
C) 4
D) 1

Q79. The length of a rectangle is increased by 20% and the breadth is decreased by 20%. What happens to the area?

A) Decreases by 4%
B) Increases by 4%
C) No change
D) Decreases by 2%

Q80. A cistern has a leak that empties it in 8 hours. A tap fills it in 6 hours. If both are open, how long to fill the empty cistern?

A) 12 hours
B) 24 hours
C) 14 hours
D) 48 hours

Q81. In how many ways can a committee of 3 be chosen from 8 people?

A) 24
B) 336
C) 112
D) 56

Q82. The average weight of 8 persons increases by 2.5 kg when a new person replaces one weighing 65 kg. What is the new person's weight?

A) 67.5 kg
B) 75 kg
C) 85 kg
D) 80 kg

Q83. A sum of ₹12,000 is lent at 10% per year compound interest. What is the interest for the third year only?

A) ₹1,200
B) ₹1,452
C) ₹1,320
D) ₹3,972

Q84. The sum of the digits of a two-digit number is 9. If the digits are reversed, the number increases by 27. What is the number?

A) 36
B) 27
C) 45
D) 63

Q85. Milk and water are in the ratio 5 : 3 in 40 litres of a mixture. How much water must be added to make the ratio 1 : 1?

A) 5 litres
B) 10 litres
C) 15 litres
D) 8 litres

Q86. What is the sum of all natural numbers from 1 to 100?

A) 5,050
B) 4,950
C) 10,100
D) 5,000

Q87. A car covers a distance in 6 hours at 60 km/h. At what speed must it travel to cover the same distance in 4 hours?

A) 80 km/h
B) 75 km/h
C) 100 km/h
D) 90 km/h

Q88. If $\log_{10} 2 = 0.3010$, what is $\log_{10} 8$?

A) 2.4080
B) 0.6020
C) 0.9030
D) 0.3010

Q89. A and B together finish a work in 8 days. A alone takes 12 days. How long does B alone take?

A) 4 days
B) 24 days
C) 16 days
D) 20 days

Q90. What is the smallest number that must be added to 1,000 to make it exactly divisible by 23?

A) 2
B) 11
C) 12
D) 13

Q91. An item is sold at successive discounts of 20% and 10%. What single discount is equal to these?

A) 28%
B) 32%
C) 30%
D) 25%

Q92. The perimeter of a square is 48 cm. What is the length of its diagonal?

A) 24 cm
B) 12 cm
C) $6\sqrt{2}$ cm
D) $12\sqrt{2}$ cm

Q93. A man rows 15 km/h in still water. The stream flows at 3 km/h. How long will he take to row 36 km upstream?

A) 4 hours
B) 2 hours
C) 2.4 hours
D) 3 hours

Q94. How many numbers from 1 to 100 are divisible by 3 or 5?

A) 50
B) 47
C) 33
D) 53

Q95. A bag contains 4 red and 6 green balls. Two balls are drawn without replacement. What is the probability that both are red?

A) 4/25
B) 2/5
C) 2/15
D) 1/5

Q96. If 20% of A equals 30% of B, what is A : B?

A) 3 : 2
B) 2 : 3
C) 1 : 1
D) 5 : 6

Q97. The volume of a cube is 512 cm^3. What is its total surface area?

A) 256 cm^2
B) 384 cm^2
C) 64 cm^2
D) 512 cm^2

Q98. A student needs 40% to pass. He gets 160 marks and fails by 20 marks. What are the maximum marks?

A) 500
B) 400
C) 360
D) 450

Q99. Two numbers are in the ratio 3 : 4 and their LCM is 84. What is the larger number?

A) 32
B) 24
C) 28
D) 21

Q100. A watch loses 5 minutes every hour. It is set right at 12 noon. What time will it show at 6 PM the same day?

A) 5:30 PM
B) 5:45 PM
C) 5:35 PM
D) 5:25 PM

ADVANCED Questions 101 – 150

Q101. A, B and C can do a work in 10, 12 and 15 days respectively. They start together, but A leaves after 2 days and B leaves 3 days before the work is finished. How many days does the whole work take?

A) 8 days
B) 7 days
C) 9 days
D) 6 days

Q102. A sum of money at compound interest amounts to ₹8,820 in 2 years and ₹9,261 in 3 years. What is the rate of interest?

A) 5%
B) 4%
C) 7.5%
D) 6%

Q103. Two trains start at the same time from A and B, 300 km apart, towards each other at 70 km/h and 80 km/h. How far from A do they meet?

A) 150 km
B) 130 km
C) 140 km
D) 160 km

Q104. In how many ways can 5 boys and 3 girls sit in a row so that no two girls sit together?

A) 2,880
B) 7,200
C) 40,320
D) 14,400

Q105. A container has 80 litres of milk. 8 litres are taken out and replaced with water. This is done 2 more times (3 times in total). How much milk is left?

A) 56 litres
B) 58.32 litres
C) 57.6 litres
D) 60 litres

Q106. What is the remainder when 2^100 is divided by 7?

A) 4
B) 3
C) 1
D) 2

Q107. A shopkeeper uses a weight of 900 g instead of 1 kg and sells at cost price. What is his profit percentage?

A) 11.11%
B) 9.09%
C) 10%
D) 12.5%

Q108. The probability that A hits a target is 1/3 and that B hits it is 1/4. If both shoot once, what is the probability that the target is hit?

A) 1/12
B) 7/12
C) 1/2
D) 5/12

Q109. A train passes a station platform in 36 seconds and a man standing on the platform in 20 seconds. If the train's speed is 54 km/h, what is the length of the platform?

A) 240 m
B) 180 m
C) 225 m
D) 300 m

Q110. A can do a job in 20 days. A and B together can do it in 12 days. They work together for 4 days, then A leaves. In how many more days will B finish the remaining work?

A) 24 days
B) 15 days
C) 16 days
D) 20 days

Q111. How many different 4-letter words (with or without meaning) can be formed from the letters of 'LOGARITHM' without repetition?

A) 126
B) 3,024
C) 6,561
D) 1,512

Q112. A person invests ₹20,000, part at 6% and the rest at 8% simple interest. The total yearly interest is ₹1,440. How much is invested at 8%?

A) ₹10,000
B) ₹8,000
C) ₹12,000
D) ₹14,000

Q113. The average of 11 results is 50. The average of the first 6 is 49 and of the last 6 is 52. What is the 6th result?

A) 56
B) 52
C) 48
D) 50

Q114. What is the number of trailing zeros in 100! (100 factorial)?

A) 20
B) 10
C) 25
D) 24

Q115. A boat takes 3 hours more to go 36 km upstream than to return downstream. If the speed of the stream is 3 km/h, what is the speed of the boat in still water?

A) 12 km/h
B) 9 km/h
C) 10 km/h
D) 6 km/h

Q116. Pipes A and B can fill a tank in 12 and 15 hours. Both are opened, but B is closed after some time and the tank is filled in 10 hours in total. After how many hours was B closed?

A) 5 hours
B) 2 hours
C) 2.5 hours
D) 3 hours

Q117. A sum doubles in 5 years at compound interest. In how many years will it become 8 times?

A) 15 years
B) 10 years
C) 40 years
D) 20 years

Q118. The ratio of two numbers is 5 : 7. If 6 is subtracted from each, the ratio becomes 1 : 2. What is the sum of the numbers?

A) 18
B) 30
C) 24
D) 36

Q119. A man buys 12 lollipops for ₹10 and sells 10 lollipops for ₹12. What is his profit percentage?

A) 24%
B) 44%
C) 20%
D) 40%

Q120. From a group of 6 men and 4 women, a committee of 5 is to be formed with at least 2 women. In how many ways can this be done?

A) 120
B) 180
C) 246
D) 186

Q121. Three bells ring at intervals of 12, 18 and 30 minutes. They all ring together at 8:00 AM. When will they next ring together?

A) 10:30 AM
B) 11:00 AM
C) 12:00 PM
D) 9:00 AM

Q122. At what time between 4 and 5 o'clock will the hands of a clock coincide?

A) 21 9/11 minutes past 4
B) 20 minutes past 4
C) 22 minutes past 4
D) 21 3/11 minutes past 4

Q123. An article is marked 50% above cost. After two successive discounts of 10% and 10%, what is the profit percentage?

A) 25%
B) 30%
C) 21.5%
D) 20%

Q124. A, B and C rent a pasture. A puts 10 oxen for 7 months, B 12 oxen for 5 months, C 15 oxen for 3 months. The rent is ₹1,750. What does C pay?

A) ₹700
B) ₹600
C) ₹500
D) ₹450

Q125. The sum of the squares of three consecutive odd numbers is 251. What is the middle number?

A) 7
B) 13
C) 11
D) 9

Q126. A bag has 5 white, 4 black and 3 red balls. Three balls are drawn at random. What is the probability that all three are of different colours?

A) 3/11
B) 5/22
C) 1/4
D) 6/55

Q127. Two vessels contain milk and water in ratios 3 : 1 and 5 : 3. In what ratio should they be mixed to get milk and water in the ratio 2 : 1?

A) 3 : 2
B) 1 : 2
C) 2 : 1
D) 1 : 1

Q128. A starts a business with ₹40,000. After 4 months B joins with ₹60,000. At the end of the year, the profit is ₹28,000. What is A's share?

A) ₹10,000
B) ₹16,000
C) ₹14,000
D) ₹12,000

Q129. A man travels 600 km partly by train and partly by car. If he covers 120 km by train and the rest by car, it takes 8 hours. If 200 km by train and the rest by car, it takes 20 minutes more. What is the speed of the train?

A) 100 km/h
B) 75 km/h
C) 60 km/h
D) 80 km/h

Q130. What is the sum of all two-digit numbers divisible by 7?

A) 728
B) 742
C) 735
D) 700

Q131. If a + b = 10 and ab = 21, what is $a^3 + b^3$?

A) 630
B) 370
C) 349
D) 1,000

Q132. A shopkeeper gives 3 articles free with every 12 bought, and still makes a 20% profit. What is the ratio of marked price to cost price per article?

A) 5 : 4
B) 6 : 5
C) 4 : 3
D) 3 : 2

Q133. Working together, A and B finish a job in 6 days. B and C finish it in 10 days, and A and C in 7.5 days. How long will A alone take?

A) 15 days
B) 10 days
C) 9 days
D) 12 days

Q134. A person walking at 4 km/h reaches his office 10 minutes late. Walking at 5 km/h, he reaches 5 minutes early. How far is his office?

A) 5 km
B) 7.5 km
C) 6 km
D) 4 km

Q135. What is the remainder when 17^23 is divided by 16?

A) 15
B) 0
C) 1
D) 7

Q136. A father is 4 times as old as his son. In 20 years he will be twice as old as his son. What is the father's present age?

A) 44 years
B) 48 years
C) 36 years
D) 40 years

Q137. A cube of side 4 cm is painted on all faces and cut into 1 cm cubes. How many small cubes have exactly two faces painted?

A) 24
B) 8
C) 16
D) 32

Q138. If the selling price is doubled, the profit triples. What is the profit percentage?

A) 66.67%
B) 120%
C) 50%
D) 100%

Q139. The compound interest on a sum for 2 years at 10% per year is ₹525. What is the simple interest on the same sum for the same period and rate?

A) ₹475
B) ₹500
C) ₹525
D) ₹550

Q140. A circular track is 600 m long. A and B start together from the same point in opposite directions at 4 m/s and 6 m/s. When will they first meet?

A) After 300 seconds
B) After 150 seconds
C) After 60 seconds
D) After 100 seconds

Q141. How many diagonals does a polygon with 12 sides have?

A) 60
B) 48
C) 54
D) 66

Q142. What is the probability of getting at least one head when 3 coins are tossed?

A) 7/8
B) 3/8
C) 1/2
D) 1/8

Q143. A sum is divided among A, B and C such that A gets ₹30 more than B, and B gets ₹40 more than C. If the total is ₹500, what does A get?

A) ₹170
B) ₹200
C) ₹230
D) ₹130

Q144. A train leaves at 6 AM at 40 km/h. Another leaves the same station at 8 AM in the same direction at 60 km/h. At what time will the second train catch the first?

A) 10 AM
B) 11 AM
C) 1 PM
D) 12 noon

Q145. The HCF of two numbers is 12 and their product is 4,320. How many such pairs of numbers exist?

A) 4
B) 3
C) 2
D) 1

Q146. The difference between a number and its 3/5 is 50. What is the number?

A) 75
B) 125
C) 150
D) 100

Q147. A man spends 75% of his income. His income rises by 20% and his spending rises by 10%. By what percent do his savings increase?

A) 25%
B) 40%
C) 50%
D) 10%

Q148. If 2^x = 32 and 3^y = 81, what is x + y?

A) 7
B) 10
C) 8
D) 9

Q149. A mixture of 60 litres has milk and water in the ratio 2 : 1. How much milk must be added to make the ratio 3 : 1?

A) 10 litres
B) 15 litres
C) 20 litres
D) 30 litres

Q150. A sum of ₹15,000 is invested at 8% per annum compounded half-yearly. What is the amount after 1 year?

A) ₹16,224
B) ₹16,000
C) ₹16,248
D) ₹16,200

PART B

Answer Key

BEGINNER Questions 1 – 50

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	D	2	B	3	A	4	C	5	A
6	B	7	D	8	C	9	A	10	C
11	B	12	D	13	D	14	A	15	B
16	C	17	C	18	A	19	D	20	B
21	A	22	C	23	B	24	D	25	D
26	C	27	B	28	A	29	A	30	D
31	B	32	C	33	D	34	B	35	A
36	C	37	A	38	C	39	B	40	D
41	D	42	B	43	C	44	A	45	A
46	C	47	B	48	D	49	C	50	A

INTERMEDIATE Questions 51 – 100

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
51	B	52	D	53	A	54	B	55	C
56	D	57	A	58	C	59	D	60	B
61	B	62	D	63	A	64	C	65	B
66	D	67	C	68	A	69	B	70	C
71	D	72	A	73	C	74	A	75	B
76	D	77	D	78	C	79	A	80	B
81	D	82	C	83	B	84	A	85	B
86	A	87	D	88	C	89	B	90	C
91	A	92	D	93	D	94	B	95	C
96	A	97	B	98	D	99	C	100	A

ADVANCED Questions 101 – 150

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
101	B	102	A	103	C	104	D	105	B
106	D	107	A	108	C	109	A	110	D
111	B	112	C	113	A	114	D	115	B
116	C	117	A	118	C	119	B	120	D
121	B	122	A	123	C	124	D	125	D
126	A	127	B	128	C	129	C	130	A

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
131	B	132	D	133	B	134	A	135	C
136	D	137	A	138	D	139	B	140	C
141	C	142	A	143	B	144	D	145	A
146	B	147	C	148	D	149	C	150	A

PART C

How to Solve — Step-by-Step Solutions

Each solution shows the correct option and a simple way to reach it. Try to understand the method, not just the answer — the same method works for many similar questions.

BEGINNER Questions 1 – 50

Q1 Answer: D) 120

25% means one quarter. 480 ÷ 4 = 120.

Q2 Answer: B) 15%

Profit = 920 − 800 = ₹120.

Profit % = (Profit ÷ Cost) × 100 = 120 ÷ 800 × 100 = 15%.

Q3 Answer: A) 24

Sum = 12 + 18 + 24 + 30 + 36 = 120. Count = 5.

Average = 120 ÷ 5 = 24.

Q4 Answer: C) 60 km/h

Speed = Distance ÷ Time = 180 ÷ 3 = 60 km/h.

Q5 Answer: A) ₹1,200

Simple Interest = P × R × T ÷ 100 = 5000 × 8 × 3 ÷ 100 = ₹1,200.

Q6 Answer: B) ₹450

Total parts = 5 + 3 = 8. One part = 720 ÷ 8 = 90.

A gets 5 parts = 5 × 90 = ₹450.

Q7 Answer: D) 36

$12 = 2^2 \times 3$ and $18 = 2 \times 3^2$. LCM takes the highest power of each prime: $2^2 \times 3^2 = 36$.

Q8 Answer: C) 24

$48 = 2^4 \times 3$ and $72 = 2^3 \times 3^2$. HCF takes the lowest power of common primes: $2^3 \times 3 = 24$.

Q9 Answer: A) ₹180

Cost of 1 pen = 96 ÷ 8 = ₹12. Cost of 15 pens = 15 × 12 = ₹180.

Q10 Answer: C) 300

Increased by 20% means 120% of the number = 360.

Number = 360 × 100 ÷ 120 = 300.

Q11 Answer: B) 125

Use $a^2 - b^2 = (a + b)(a - b) = (15 + 10)(15 - 10) = 25 \times 5 = 125$.

Q12 Answer: D) 1/10

If the whole job takes 10 days, each day A does 1/10 of it.

Q13 Answer: D) 20%

Decrease = 250 − 200 = 50. Percentage decrease = 50 ÷ 250 × 100 = 20% (always divide by the original value).

Q14 Answer: A) 150

250 ÷ 5 = 50, and 3 × 50 = 150.

Q15 Answer: B) 42

Differences are 4, 6, 8, 10, so the next difference is 12. 30 + 12 = 42.

Q16 Answer: C) 30

Larger = (Sum + Difference) ÷ 2 = (50 + 10) ÷ 2 = 30.

Q17 Answer: C) 210

1 hour = 60 minutes. 3.5 × 60 = 210 minutes.

Q18 Answer: A) 20 m/s

To pass a pole the train covers only its own length.

Speed = 120 ÷ 6 = 20 m/s.

Q19 Answer: D) 20 m/s

Multiply km/h by 5/18 to get m/s. 72 × 5 ÷ 18 = 20 m/s.

Q20 Answer: B) 1/4

0.25 = 25/100. Divide top and bottom by 25 → 1/4.

Q21 Answer: A) ₹360

Loss = 10% of 400 = ₹40. Selling price = 400 − 40 = ₹360.

Q22 Answer: C) 12

3 parts = 18, so 1 part = 6. Girls = 2 parts = 12.

Q23 Answer: B) 36

36 × 36 = 1,296. (Hint: the number ends in 6, so the root ends in 4 or 6; 35^2 = 1,225 is close, so try 36.)

Q24 Answer: D) 29

39 = 3 × 13, 51 = 3 × 17 and 57 = 3 × 19. Only 29 has no factor other than 1 and itself.

Q25 Answer: D) 12.5 km

2 hours 30 minutes = 2.5 hours. Distance = 5 × 2.5 = 12.5 km.

Q26 Answer: C) 80 years

Total = Average × Count = 20 × 4 = 80 years.

Q27 Answer: B) 120

60% of 500 = 300. Then 40% of 300 = 120.

Q28 Answer: A) 24

x = 19 − 7 = 12, so 2x = 24.

Q29 Answer: A) 7

Divide 50 by 7: 50 ÷ 7 = 7 remainder 1. So the multiples are 7, 14, ..., 49 → 7 numbers.

Q30 Answer: D) 40 cm

Perimeter = 2 × (Length + Breadth) = 2 × (12 + 8) = 40 cm.

Q31 Answer: B) 81 sq cm

Area of a square = side × side = 9 × 9 = 81 sq cm.

Q32 Answer: C) 10%

Doubling means interest = principal (100%). 100% earned in 10 years → 10% per year.

Q33 Answer: D) 11

$\sqrt{64} = 8$ and $\sqrt[3]{27} = 3$. 8 + 3 = 11.

Q34 Answer: B) 6

Work = 5 × 12 = 60 worker-days. With 10 workers: 60 ÷ 10 = 6 days. (More workers → fewer days.)

Q35 Answer: A) 1

Use LCM 6: 3/6 + 2/6 + 1/6 = 6/6 = 1.

Q36 Answer: C) 3/8

Probability = favourable ÷ total = 3 ÷ (3 + 5) = 3/8.

Q37 Answer: A) 0.405

Write all with three decimals: 0.500, 0.450, 0.405, 0.540. The smallest is 0.405.

Q38 Answer: C) ₹600

Selling price = 110% of cost. Cost = 660 × 100 ÷ 110 = ₹600.

Q39 Answer: B) 1/3

In 1 hour it fills 1/6. In 2 hours it fills 2/6 = 1/3.

Q40 Answer: D) 54

Follow BODMAS: multiply and divide first. 7 × 8 = 56, 12 ÷ 4 = 3. Then 56 + 3 − 5 = 54.

Q41 Answer: D) 75%

45 ÷ 60 × 100 = 75%.

Q42 Answer: B) 25

Middle number = 72 ÷ 3 = 24. The numbers are 23, 24, 25, so the largest is 25.

Q43 Answer: C) 86,400

24 hours × 60 minutes × 60 seconds = 86,400 seconds.

Q44 **Answer: A) 2**

7 × 14 = 98. 100 − 98 = 2.

Q45 **Answer: A) ₹4,500**

15% of 30,000 = 30,000 × 15 ÷ 100 = ₹4,500.

Q46 **Answer: C) ₹4**

1 dozen = 12. ₹48 ÷ 12 = ₹4.

Q47 **Answer: B) 243**

Each number is multiplied by 3. 81 × 3 = 243.

Q48 **Answer: D) 90°**

Angles add up to 180°. Total parts = 6, one part = 30°. Largest = 3 × 30 = 90°.

Q49 **Answer: C) 37.5%**

3 ÷ 8 = 0.375. Multiply by 100 → 37.5%.

Q50 **Answer: A) 3 hours 30 minutes**

From 9:45 to 12:45 is 3 hours, and 12:45 to 1:15 is 30 minutes. Total = 3 hours 30 minutes.

INTERMEDIATE Questions 51 – 100

Q51 Answer: B) 7.2 days

A's 1-day work = 1/12, B's = 1/18. Together = 1/12 + 1/18 = 3/36 + 2/36 = 5/36.

Days = 36 ÷ 5 = 7.2 days.

Q52 Answer: D) 72 km/h

To cross a platform the train covers train length + platform length = 500 m.

Speed = 500 ÷ 25 = 20 m/s. In km/h: 20 × 18 ÷ 5 = 72 km/h.

Q53 Answer: A) ₹2,100

Amount = 10,000 × 1.1 × 1.1 = ₹12,100. CI = 12,100 − 10,000 = ₹2,100.

(Shortcut: CI for 2 years = SI + interest on interest = 2,000 + 100.)

Q54 Answer: B) 12%

Let cost = 100. Marked price = 140. After 20% discount: 140 × 0.8 = 112.

Profit = 12 on 100 → 12%.

Q55 Answer: C) 48

Total of 5 numbers = 5 × 40 = 200. Total of 4 numbers = 4 × 38 = 152.

Removed number = 200 − 152 = 48.

Q56 Answer: D) 3 km/h

Downstream speed = 24 ÷ 2 = 12 km/h. Upstream speed = 24 ÷ 4 = 6 km/h.

Stream speed = (12 − 6) ÷ 2 = 3 km/h.

Q57 Answer: A) 5 : 6

Let salaries be 400 and 500. New A = 400 × 1.25 = 500. New B = 500 × 1.2 = 600.

New ratio = 500 : 600 = 5 : 6.

Q58 Answer: C) 60 minutes

Per minute: +1/20 + 1/30 − 1/15 = 3/60 + 2/60 − 4/60 = 1/60.

So the tank fills in 60 minutes.

Q59 Answer: D) 60

APPLE has 5 letters with P repeated twice.

Arrangements = 5! ÷ 2! = 120 ÷ 2 = 60.

Q60 Answer: B) ₹5,700

Interest for 2 years (from year 3 to 5) = 7,200 − 6,600 = 600, so 1 year = 300.

Interest for 3 years = 900. Principal = 6,600 − 900 = ₹5,700.

Q61 Answer: B) 5/36

Pairs giving 8: (2,6), (3,5), (4,4), (5,3), (6,2) → 5 outcomes out of 36.

Probability = 5/36.

Q62 Answer: D) 2 : 1

Use alligation: (55 − 45) : (45 − 40) = 10 : 5 = 2 : 1 (cheaper : dearer).

Q63 Answer: A) 48 km/h

For equal distances, average speed = 2xy ÷ (x + y) = 2 × 40 × 60 ÷ 100 = 48 km/h. (It is not the simple average 50.)

Q64 Answer: C) 60,500

After 1 year: 50,000 × 1.1 = 55,000. After 2 years: 55,000 × 1.1 = 60,500.

Q65 Answer: B) 7

Unit digits of powers of 7 repeat every 4: 7, 9, 3, 1.

45 ÷ 4 leaves remainder 1, so the unit digit is the 1st in the cycle → 7.

Q66 Answer: D) 20%

Reduction % = r ÷ (100 + r) × 100 = 25 ÷ 125 × 100 = 20%.

Q67 Answer: C) ₹9,000

Ratio of investment = 30,000 : 45,000 = 2 : 3.

B's share = 3/5 × 15,000 = ₹9,000.

Q68 Answer: A) 130°

Angle = |30H − 5.5M| = |30 × 3 − 5.5 × 40| = |90 − 220| = 130°.

Q69 Answer: B) ₹10,000

For 2 years, CI − SI = P × $(R/100)^2$.

25 = P × $(5/100)^2$ = P/400, so P = ₹10,000.

Q70 Answer: C) 1% loss

When the same selling price gives x% gain and x% loss, there is always a loss of $x^2/100$ % = 100/100 = 1% loss.

Q71 Answer: D) 60

First place: 5 choices, second: 4, third: 3. Total = 5 × 4 × 3 = 60.

Q72 Answer: A) 240 m

Relative speed (same direction) = 54 − 6 = 48 km/h = 48 × 5/18 = 40/3 m/s.

Length = 40/3 × 18 = 240 m.

Q73 Answer: C) 18 years

Let ages be 3x and 5x. (3x + 6) : (5x + 6) = 2 : 3 → 9x + 18 = 10x + 12 → x = 6.

A = 3 × 6 = 18 years.

Q74 Answer: A) 20 days

Use $M_1D_1H_1 = M_2D_2H_2$: 15 × 20 × 8 = 20 × D × 6 → 2400 = 120D → D = 20 days.

Q75 Answer: B) 29

LCM of 6, 8, 12 = 24. Add the remainder: 24 + 5 = 29.

Q76 Answer: D) ₹400

A's cost × 1.2 × 1.25 = 600 → A's cost × 1.5 = 600 → ₹400.

Q77 Answer: D) 10

Like at least one = 60 + 50 − 20 = 90. Neither = 100 − 90 = 10.

Q78 Answer: C) 4

Number = 5k + 3. Square leaves the same remainder as $3^2 = 9$. 9 ÷ 5 leaves remainder 4.

Q79 Answer: A) Decreases by 4%

New area factor = 1.2 × 0.8 = 0.96, so the area falls by 4%.

Q80 Answer: B) 24 hours

Per hour: 1/6 − 1/8 = 4/24 − 3/24 = 1/24. So it fills in 24 hours.

Q81 Answer: D) 56

Order does not matter, so use combinations: 8C3 = (8 × 7 × 6) ÷ (3 × 2 × 1) = 56.

Q82 Answer: C) 85 kg

Total weight goes up by 8 × 2.5 = 20 kg.

New person = 65 + 20 = 85 kg.

Q83 Answer: B) ₹1,452

Amount after 2 years = 12,000 × 1.21 = 14,520. Interest in the 3rd year = 10% of 14,520 = ₹1,452.

Q84 Answer: A) 36

Let digits be a (tens) and b (units). a + b = 9 and (10b + a) − (10a + b) = 27 → b − a = 3.

So b = 6, a = 3 → number 36.

Q85 Answer: B) 10 litres

Milk = 25 L, water = 15 L. For 1 : 1 water must also be 25 L. Add 25 − 15 = 10 litres.

Q86 Answer: A) 5,050

Sum = n(n + 1) ÷ 2 = 100 × 101 ÷ 2 = 5,050.

Q87 Answer: D) 90 km/h

Distance = 60 × 6 = 360 km. Speed needed = 360 ÷ 4 = 90 km/h.

Q88 Answer: C) 0.9030

$8 = 2^3$, so log 8 = 3 × log 2 = 3 × 0.3010 = 0.9030.

Q89 Answer: B) 24 days

B's 1-day work = 1/8 − 1/12 = 3/24 − 2/24 = 1/24. So B takes 24 days.

Q90 Answer: C) 12

23 × 43 = 989, remainder = 1000 − 989 = 11. Add 23 − 11 = 12 (1,012 = 23 × 44).

Q91 **Answer: A) 28%**

Price factor = 0.8 × 0.9 = 0.72. Single discount = 100 − 72 = 28%.

Q92 **Answer: D) 12√2 cm**

Side = 48 ÷ 4 = 12 cm. Diagonal = side × √2 = 12√2 cm.

Q93 **Answer: D) 3 hours**

Upstream speed = 15 − 3 = 12 km/h. Time = 36 ÷ 12 = 3 hours.

Q94 **Answer: B) 47**

Divisible by 3: 33. By 5: 20. By both (by 15): 6.

By 3 or 5 = 33 + 20 − 6 = 47.

Q95 **Answer: C) 2/15**

P = (4/10) × (3/9) = 12/90 = 2/15.

Q96 **Answer: A) 3 : 2**

0.2A = 0.3B → A/B = 0.3/0.2 = 3/2. So A : B = 3 : 2.

Q97 **Answer: B) 384 cm²**

Side = $\sqrt[3]{512}$ = 8 cm. Surface area = 6 × 8² = 384 cm².

Q98 **Answer: D) 450**

Pass marks = 160 + 20 = 180. 40% of maximum = 180 → maximum = 180 ÷ 0.4 = 450.

Q99 **Answer: C) 28**

Numbers are 3x and 4x, LCM = 12x = 84 → x = 7. Larger = 4 × 7 = 28.

Q100 **Answer: A) 5:30 PM**

In 6 hours it loses 6 × 5 = 30 minutes. So it shows 6:00 − 0:30 = 5:30 PM.

ADVANCED Questions 101 – 150

Q101 Answer: B) 7 days

Let the total time be T days. A works 2 days, B works (T − 3) days, C works T days.

2/10 + (T − 3)/12 + T/15 = 1. Multiply by 60: 12 + 5(T − 3) + 4T = 60 → 9T − 3 = 60 → T = 7 days.

Q102 Answer: A) 5%

Interest in year 3 = 9,261 − 8,820 = 441, which is the interest on 8,820 for 1 year.

Rate = 441 ÷ 8,820 × 100 = 5%.

Q103 Answer: C) 140 km

Relative speed = 150 km/h, so they meet after 300 ÷ 150 = 2 hours.

Distance from A = 70 × 2 = 140 km.

Q104 Answer: D) 14,400

Seat the boys first: 5! = 120 ways. This creates 6 gaps. Choose and arrange 3 girls in those gaps: 6P3 = 120.

Total = 120 × 120 = 14,400.

Q105 Answer: B) 58.32 litres

Milk left = $80 \times (1 - 8/80)^3 = 80 \times 0.9^3 = 80 \times 0.729 = 58.32$ litres.

Q106 Answer: D) 2

Powers of 2 mod 7 repeat every 3: 2, 4, 1.

100 ÷ 3 leaves remainder 1, so 2^100 mod 7 = 2.

Q107 Answer: A) 11.11%

He gives 900 g but charges for 1000 g. Profit = 100 g on 900 g.

Profit % = 100 ÷ 900 × 100 = 11.11%.

Q108 Answer: C) 1/2

P(target missed by both) = (2/3) × (3/4) = 1/2.

P(hit at least once) = 1 − 1/2 = 1/2.

Q109 Answer: A) 240 m

Speed = 54 × 5/18 = 15 m/s. Train length = 15 × 20 = 300 m.

Train + platform = 15 × 36 = 540 m. Platform = 540 − 300 = 240 m.

Q110 Answer: D) 20 days

B alone: 1/12 − 1/20 = 5/60 − 3/60 = 1/30 → 30 days.

In 4 days together they do 4/12 = 1/3. Remaining 2/3 by B = 2/3 × 30 = 20 days.

Q111 Answer: B) 3,024

'LOGARITHM' has 9 different letters. Arrangements of 4 out of 9 = 9P4 = 9 × 8 × 7 × 6 = 3,024.

Q112 Answer: C) ₹12,000

Let x be at 8%. Then 0.08x + 0.06(20,000 − x) = 1,440 → 0.02x + 1,200 = 1,440 → x = ₹12,000.

Q113 **Answer: A) 56**

Sum of all 11 = 550. First 6 = 294, last 6 = 312. The 6th result is counted twice.

6th = 294 + 312 − 550 = 56.

Q114 **Answer: D) 24**

Count factors of 5: 100 ÷ 5 = 20, 100 ÷ 25 = 4, 100 ÷ 125 = 0. Total = 24 trailing zeros.

Q115 **Answer: B) 9 km/h**

$36/(b-3) - 36/(b+3) = 3$. Simplify: $36 \times 6 = 3(b^2 - 9) \rightarrow 216 = 3b^2 - 27 \rightarrow b^2 = 81 \rightarrow b = 9$ km/h.

Q116 **Answer: C) 2.5 hours**

A works the full 10 hours: 10/12 = 5/6 of the tank. B fills the remaining 1/6.

B's time = 1/6 × 15 = 2.5 hours.

Q117 **Answer: A) 15 years**

$8 = 2^3$. Every 5 years it doubles, so it needs 3 doublings = 15 years.

Q118 **Answer: C) 24**

$(5x - 6)/(7x - 6) = 1/2 \rightarrow 10x - 12 = 7x - 6 \rightarrow x = 2$. Numbers 10 and 14, sum = 24.

Q119 **Answer: B) 44%**

Cost of 1 = 10/12, selling price of 1 = 12/10.

Profit % = (1.2 ÷ 0.8333 − 1) × 100 = (1.44 − 1) × 100 = 44%.

Q120 **Answer: D) 186**

2W+3M: $4C2 \times 6C3 = 6 \times 20 = 120$. 3W+2M: $4C3 \times 6C2 = 4 \times 15 = 60$. 4W+1M: $1 \times 6 = 6$.

Total = 120 + 60 + 6 = 186.

Q121 **Answer: B) 11:00 AM**

LCM of 12, 18, 30 = 180 minutes = 3 hours. 8:00 AM + 3 hours = 11:00 AM.

Q122 **Answer: A) 21 9/11 minutes past 4**

The hands coincide when 30H = 5.5M. 30 × 4 = 5.5M → M = 120/5.5 = 21 9/11 minutes past 4.

Q123 **Answer: C) 21.5%**

Let cost = 100, marked = 150. After discounts: 150 × 0.9 × 0.9 = 121.5. Profit = 21.5%.

Q124 **Answer: D) ₹450**

Ox-months: A = 70, B = 60, C = 45. Total = 175.

C pays 45/175 × 1,750 = ₹450.

Q125 **Answer: D) 9**

Try 7, 9, 11: 49 + 81 + 121 = 251. The middle number is 9.

Q126 **Answer: A) 3/11**

Favourable = 5 × 4 × 3 = 60. Total = 12C3 = 220.

Probability = 60/220 = 3/11.

Q127 **Answer: B) 1 : 2**

Milk share in vessel 1 = 3/4 = 18/24, in vessel 2 = 5/8 = 15/24, and in the target mix = 2/3 = 16/24.

Alligation: Vessel 1 : Vessel 2 = (16 − 15) : (18 − 16) = 1 : 2.

Q128 **Answer: C) ₹14,000**

Profit is shared in the ratio of (money × months).

A: 40,000 × 12 = 4,80,000. B: 60,000 × 8 = 4,80,000. Ratio = 1 : 1, so A gets half of 28,000 = ₹14,000.

Q129 **Answer: C) 60 km/h**

Let train speed t, car speed c. 120/t + 480/c = 8 and 200/t + 400/c = 25/3.

Let u = 1/t, v = 1/c: 120u + 480v = 8 and 200u + 400v = 25/3. Solving gives u = 1/60, v = 1/80.

Train = 60 km/h (car = 80 km/h).

Q130 **Answer: A) 728**

Two-digit multiples of 7: 14, 21, ..., 98 → 13 terms.

Sum = 13 × (14 + 98) ÷ 2 = 13 × 56 = 728.

Q131 **Answer: B) 370**

$a^3 + b^3 = (a + b)^3 - 3ab(a + b) = 1,000 - 3 \times 21 \times 10 = 1,000 - 630 = 370$.

Q132 **Answer: D) 3 : 2**

He gives 15 articles for the price of 12. Selling 15 at 20% profit means revenue = 15 × 1.2 C = 18C.

Revenue = 12 × M. So M = 18C/12 = 1.5C → M : C = 3 : 2.

Q133 **Answer: B) 10 days**

Add all: 2(A + B + C) = 1/6 + 1/10 + 2/15 = 5/30 + 3/30 + 4/30 = 12/30 → A + B + C = 1/5.

A = 1/5 − (B + C) = 1/5 − 1/10 = 1/10 → 10 days.

Q134 **Answer: A) 5 km**

Time difference = 15 minutes = 1/4 hour. d/4 − d/5 = 1/4 → d/20 = 1/4 → d = 5 km.

Q135 **Answer: C) 1**

17 = 16 + 1, so 17 leaves remainder 1 when divided by 16. Any power of it also leaves remainder 1.

Q136 **Answer: D) 40 years**

Son = x, father = 4x. 4x + 20 = 2(x + 20) → 2x = 20 → x = 10. Father = 40 years.

Q137 **Answer: A) 24**

Cubes with two painted faces lie on edges (not corners). Each edge has (4 − 2) = 2 such cubes, and a cube has 12 edges.

12 × 2 = 24.

Q138 **Answer: D) 100%**

Let cost C and profit P: S = C + P, and 2S = C + 3P. Subtract: S = 2P, so C + P = 2P → C = P.

Profit % = P/C × 100 = 100%.

Q139 Answer: B) ₹500

CI factor for 2 years at 10% = 1.21 − 1 = 0.21. So P = 525 ÷ 0.21 = 2,500.

SI = 2,500 × 10 × 2 ÷ 100 = ₹500.

Q140 Answer: C) After 60 seconds

Opposite directions → relative speed = 4 + 6 = 10 m/s. They meet when together they cover 600 m: 600 ÷ 10 = 60 seconds.

Q141 Answer: C) 54

Diagonals = n(n − 3) ÷ 2 = 12 × 9 ÷ 2 = 54.

Q142 Answer: A) 7/8

P(no head) = $(1/2)^3$ = 1/8. P(at least one head) = 1 − 1/8 = 7/8.

Q143 Answer: B) ₹200

Let C = x. B = x + 40, A = x + 70. Total = 3x + 110 = 500 → x = 130.

A = 130 + 70 = ₹200.

Q144 Answer: D) 12 noon

By 8 AM the first train is 80 km ahead. Gap closes at 60 − 40 = 20 km/h → 4 hours. 8 AM + 4 = 12 noon.

Q145 Answer: A) 4

Write the numbers as 12a and 12b where a and b have no common factor. 144ab = 4,320 → ab = 30.

Co-prime pairs with product 30: (1, 30), (2, 15), (3, 10), (5, 6) → 4 pairs.

Q146 Answer: B) 125

x − 3x/5 = 2x/5 = 50 → x = 125.

Q147 Answer: C) 50%

Income 100, spending 75, savings 25. New income 120, new spending 82.5, new savings 37.5.

Increase = 12.5 on 25 = 50%.

Q148 Answer: D) 9

$32 = 2^5$ so x = 5. $81 = 3^4$ so y = 4. x + y = 9.

Q149 Answer: C) 20 litres

Milk = 40 L, water = 20 L. For 3 : 1, milk must be 3 × 20 = 60 L. Add 60 − 40 = 20 litres.

Q150 Answer: A) ₹16,224

Half-yearly means rate = 8 ÷ 2 = 4% per half-year, and 2 half-years in 1 year.

Amount = 15,000 × 1.04 × 1.04 = 15,000 × 1.0816 = ₹16,224.